

ANISH DHAR

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PROFESSIONAL SUMMARY

Mechanical engineer with hands-on experience in capital equipment design, structural and thermal simulation, precision machining, and cross-functional product development. Took full design ownership of a INR 3.5 Cr total-containment processing system at Glatt Systems, coordinating from layout through manufacturing release. Proficient in Autodesk Inventor, SolidWorks, ANSYS FEA/CFD, GD&T, and additive manufacturing. Track record of designing and testing mechanical assemblies under real production and competition constraints.

EXPERIENCE

Engineering Trainee – New Product Development | Glatt Systems Pvt. Ltd., Pune, IN

Jun 2025 – Present

- Took complete design ownership of a total-containment bin-washing system for potent API processing, valued at INR 3.5 Cr; delivered full design and drawing package from scratch; coordinating procurement and manufacturing.
- Designed a novel SS316 ducting joining system to eliminate legacy flanged connections; prototype successfully qualified at 2 bar pressure in a single-connection acceptance test.
- Contributed structural design to a 3-story all-steel vertical granulation line for a global pharmaceutical OEM, coordinating across mechanical, process, and controls sub-teams through pre-production validation.
- Designed external fascia panels and enclosure geometry for a new lab-scale extruder/spheronizer machine – first-of-kind development at this scale; took concept through to fabricated prototype, covering sheet metal layout, fixturing, and surface finish requirements.
- Managed engineering change notices (ECNs) across active design packages; tracked revision-controlled drawing releases through Vault PDM, liaising with vendors and shop floor.
- Incorporated shop-floor feedback into design revisions across assemblies of 100+ parts; maintained GD&T drawing standards and tolerance stack-up documentation throughout fabrication and inspection stages.

Engineering Intern – Thermal Systems | Saarloha Advanced Materials Pvt. Ltd., Pune, IN

Jun 2024 – Aug 2024

- Surveyed the water-cooling circuit of a 35T Electric Arc Furnace and continuous casting line; identified thermal loss pathways from slag deposition on cooling panels by analyzing 30+ research papers and plant instrumentation data.
- Delivered a structured improvement report with design modifications to reduce parasitic heat losses in the EAF water-cooling loop, aligned with the site's green manufacturing targets.

ENGINEERING PROJECTS & COMPETITIONS

Thermal Systems, Electronics – SAE e-BAJA & ATVC

VJTI Racing | Jun 2022 – Mar 2024

- Designed a jet-impingement cooling enclosure for the motor controller using SLA/FDM additive manufacturing (mass: 500 g); targeted a 25 °C reduction in peak operating temperature versus the baseline air-cooled configuration.
- Developed low-loss manifold geometry to reduce static pressure drop across the forced-air distribution network, decreasing auxiliary battery load; team achieved AIR 1 in Sales Presentation and AIR 5 Overall at SAE e-BAJA 2024.
- Led low- and high-voltage system design for ISIE IKR 2023 EV entry; implemented a multi-layer startup interlock sequence — awarded Best Accelerating Vehicle, Best Business Proposal, AIR 5 EV category.

Converging-Diverging Nozzle – IPA/Air Rocket Engine

Personal Project

- Designed a C-D nozzle for 300 N sea-level thrust; performed CNC machining of mild steel nozzle on a Jyoti VMC850, including fixturing design, toolpath programming, and post-test failure analysis.

Numerical Investigation of Lattice-Based Cooling Structures – Capstone

VJTI, Apr 2025

- Developed and meshed lattice unit-cell geometries in ANSYS Fluent; conducted mesh-convergence studies to characterize convective heat transfer enhancement and pressure-drop trade-offs across candidate lattice topologies for high-performance thermal management.

Compressive Behavior of Vertically Graded Lattice Structures – Honours

VJTI, May 2025

- Modelled graded-density lattice structures under uniaxial compression in ANSYS Mechanical; extracted load-displacement and stress-strain curves via structural FEA to quantify energy absorption relative to uniform-density architectures.

EDUCATION

B.Tech – Mechanical Engineering | VJTI Mumbai | CGPA: 7.67

Dec 2021 – May 2025

Relevant coursework: Thermodynamics, Fluid Mechanics & CFD, Material Sciences, Manufacturing, Solid Mechanics, Additive Manufacturing (Honours).

LICENSES & CERTIFICATIONS

Applied Computational Fluid Dynamics — Siemens

Apr 2026

Cleanroom Fundamentals & Semiconductor Technologies — University at Buffalo

Apr 2026

SOLIDWORKS: Design Study and Optimization — LinkedIn Learning

Jun 2023

TECHNICAL SKILLS

CAD / PDM: Autodesk Inventor (large assemblies, sheet metal, weldments, GD&T), SolidWorks, Vault PDM, Fusion 360

Simulation: ANSYS Mechanical (structural FEA, thermal; hex/tet meshing, convergence), ANSYS Fluent (CFD), MATLAB/Simulink

Manufacturing: CNC Machining (Jyoti VMC850), FDM & SLA additive manufacturing, SS316 & mild steel fabrication, fixturing, GD&T, tolerance stack-up, ECN/change management, DFX, thermodynamics, flow analysis

Computing: Python (SPH simulation, PINN, FEA post-processing), MATLAB (CFD, heat transfer), GitHub